Name: ______________________________ **Date:** _________________ **Class:** _______________

MyDesign Portfolio

Element H Initial Template

Element H: Prototype Testing and Data Collection Plan

Testing Details *(H1, H2, H3, H4)*

	requirement	testing setup and procedure	test cases and number of trials	expected range of results	pass/fail criteria, goal
1	Create functioning fire detection and suppression systems	Detection/Notification system: 1) Install the detection system onto the roof of the house temporarily 2) Hold a lighter/flame near the system to see if it works 3) Check if the system starts beeping 4) 1) Hold the prototype in a hand using the grip in this photo.	Detection system: Hold lighter near alarm and wait for beeping (two-three times)	The detection system may: -beep or not beep -not detect the fire -not be connected to the computer -not be close enough to the fire	**Detection system:** **Fail**: fire is not detected/notified (beeping) **Pass**: the alarm beeps because a fire is detected and notifies the "occupants" **Goal**: all parts of the prototype remain unchanged when the prototype is returned to its original position
2	Create a functioning suppression system	Suppression System: 1) Create plastic bags using plastic bags that were ripped apart 2) Fill them with water to check for leaks 3) Seal the leaks 4) Refill the bag with water	Suppression System: Fill and refill bags with water (four times). Tie the bag to the ceiling and check that it doesn't come	The suppression system may: -leak -not hold onto the ceiling	**Suppression System:** **Fail:** water balloons burst or water leaks out **Pass:** the water balloons pop when

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		5) Make sure it hangs to the ceiling (required refastening the "sprinkler system")	off (5-6 times)	-not be big enough to hold necessary amount of water	the plastic catches fire, and the water inside suppresses the fire. **Goal:** the balloon opens and the water puts out the fire
3	Build a 18x12 inch apartment with a long glass wall, all other walls, floor, and ceiling must be solid, opaque materials, two rooms (12x12 and 6x12), a removable ceiling, one doorway (6.5x2.5) leading out of the front room to outside, One doorway (6.5x 2.5) connecting the rooms (interior), one window on two exterior walls for	1) "Shake" the different parts of the house to make sure they are properly attached and won't fall off 2) Double check the measurements and requirements for glass wall, doors, and number of windows 3) Reattach any parts of the house that are not fastened properly	Shook the different parts of the house (multiple times) Rechecked all the wall, door, and window measurements (2 times) Reattached loose pieces (9-10 times)	Some parts of the house might fall off or they might all stay attached. Some pieces might be missing or we might have all of the pieces but need to fix the location of the pieces.	**Fail:** The house falls apart or some pieces are loose and cannot be reattached for some reason. **Pass:** The pieces which are loose can be reattached and all the pieces/criteria are present. **Goal:** None of the pieces fall off and everything is in the required position.

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	the larger room (two windows) Windows should be at least 3" wide and 2" high)				
4	Create realistic mini furniture in which can catch fire. There must at least be 2 Beds (6 X 3 x 2), 2 Desks (4 x 2 x 3), 2 chairs (2.5 x 2.5 x 2) and other flammable decor like curtains, carpet, etc.	1) Put all of the furniture standing up right to make sure they are stable 2) Check to make sure measurements meet criteria	Placed furniture standing Measured furniture (once)	Some pieces of furniture might fall apart or stay together The measurements could be off or correct	**Fail:** The furniture falls apart/breaks or some pieces are loose Measurements do not meet requirement **Pass:** Furniture does not fall apart/break and no pieces are loose Measurements meet requirement **Goal:** Furniture does not fall apart/break and no pieces are loose Measurements meet requirement

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Feedback From Field Experts *(H5)*

Name of your field expert:

Dr. Jim Milke

Why are they a field expert?

Dr. Milke is a field expert because he works as a professor in the University of Maryland's Fire Protection Engineering Department. He holds a B.S. in Fire Protection Engineering and a M.S. in Mechanical Engineering, as well as a Ph.D. in Aerospace Engineering, all from the University of Maryland, whose engineering department organized the Burn Day challenge. Therefore, Dr. Milke is knowledgeable in the field of our project and can provide feedback and information regarding fire detection and suppression systems.

What did they say to you about the tests you are proposing?

Dr. Milke told us after the project was complete that while our detection system (fire alarm) works well, our suppression system (sprinklers) should have been much bigger in order to effectively suppress the fire since we did not have enough water in our system.

Revised Testing Details That Incorporate Field Expert Feedback *(H5)*

Since the feedback from our field expert was given after our project was complete, we could not implement it in our project. However, if we were to make revisions, we would have fixed the sprinkler system, filling the balloons with a lot of water, and also making sure it was fastened well to the ceiling.

Element H: Prototype Testing and Data Collection Plan						
	5	4	3	2	1	0
1. Addressing Requirements. The final prototype Testing Plan addresses ______________ of the *high priority* design requirements.	*all or nearly all*	*many*	*some*	*a few*	*one*	*none - or is missing altogether*
2. Conduct of Tests. The Testing Plan ______________ in describing the conduct of tests (through physical and/or mathematical modeling).	*is effective*	*is generally effective*	*is only adequate*	*is partially effective*	*is at least minimally effective*	*is void*
3. Quantity of Tests Planned. The Testing Plan defines ______________________ of *all* design requirements based on the instructional context.	sufficient quantity to feasibly test *100%*	sufficient quantity to feasibly test *80%*	sufficient quantity to feasibly test *60%*	sufficient quantity to feasibly test *40%*	*sufficient quantity to feasibly test 20%*	*no feasible test*
4. Logic of the Plan. The Testing Plan provides ______________________.	a logical and *well -developed* explanation	*a logical and generally developed* explanation	*a generally logical and adequately developed* explanation	*a somewhat logical and partially-developed* explanation	*at least a generally logical and/or partially -developed explanation*	*no explanation*

5. Use of Field Experts. The Testing Plan is confirmed by _____________ .	*more than two* field experts with feedback provided and incorporated	*two* field experts with feedback provided and incorporated	one field expert with feedback provide*d and incorporated*	one field expert *with feedback provided*	*one field expert but no response was documented*	*Field expertise is not mentioned or addressed and field experts are not consulted.*